EPSR-test-ex64 rings: water H-bond rings, two distributions

Program(s): rings | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `rings` [1], which counts closed rings of bonded molecules with the shortest-path criterion of Franzblau [2] (as cited in `chains_sp_criterion.f`): a ring is accepted only if no shorter path through the network connects two of its members. Bonds are defined as in `chains` (three distance windows and an angle), and histograms of neighbour numbers and ring sizes are written.

Code exercised

Fortran (original, fortran_src/):

- `rings_sp_criterion.f`: `run_rings`, `find_rings` (shortest-path ring search, 'no short cut' test)
- `chains_routines.f`: `match`, `shortest_path`
- `read_and_save_files.f`: `outgofz`, `save_files_others`; `setup_symbols_generic.f`: `setup_symbols_rings` (computed `nsize` default)

Rust (port, rust/epsr-core/src/):

- `tools/rings.rs`: `run`, `find_rings`
- `tools/chains.rs`: `run_chains_or_rings`, `read_criteria`, `find_bonds`, `shortest_path`, `fmatch`
- `tools/clusters.rs`: `outgofz`; `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `RINGS`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å³), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [3]. Distribution 1: water hydrogen bonds O1-H1···O1 (O-H 0.8-1.2, O···O 2.4-3.4, H···O 1.4-2.5 Å, angle <= 40°); distribution 2: two criteria `O1 H1 O*` (water or DMSO acceptor) and `O1 H1 S`. Ring statistics characterise the topology of the hydrogen-bond network; the README reports 41 and 90 rings found.

Inputs

`hbring.RINGS.dat`: `nsize 12` (so `nspmax` = 7), `nsumt 0`, two distributions. `200DMSO400Water.ato` copied unchanged from EPSR-test; the keyword file was written by hand (the Fortran program creates no template when the named file does not exist; keywords and defaults from `setup_symbols_generic.f`).

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`repeat 2` with `nsumt 0` (reference after pass 2: `nsumt 2`). Exercises the shortest-path ring search with the 'no short cut' test (`nspvaluet` route counts), the shared `spmin` bookkeeping and both histograms. Compared after each pass: `hbring.RINGS.n01`, the rewritten `hbring.RINGS.dat` and `fort.14` (means and spreads of the histograms). The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] D. S. Franzblau, "Computation of ring statistics for network models of solids", Phys. Rev. B 44, 4925-4930 (1991).
[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.